

NIMS Emergency Department App — Consolidated Change Specification

Version: 1.0 · **Compiled:** 12 September 2026 **Applies to:** `app/` — NIMS ED Register (React + Vite + IndexedDB/Supabase) **Status:** consolidated from all stakeholder inputs; sections marked ▲ need a clinical decision before build.

1. What this document is

Every requested change to the app, gathered from the four separate input streams that arrived in `docs/`, de-duplicated, reconciled where they disagreed, and mapped onto the code that exists today.

Source inventory

Code	Source	Character
SPEC	<code>nims-ed-app-spec.docx</code>	Typed summary — the most organised source, but not the most complete
N1 – N4	Handwritten notes, pages 1–4 (<code>Pasted image (7) – (10).png</code>)	Original clinical notes from the meeting
CHAT	WhatsApp forwards (<code>Pasted image.png</code> , (2) – (6) .png)	Point-by-point dictation; contains items absent from SPEC
IDEAS	<code>duplicate ideas evidence.txt</code>	Earlier brainstorm — platform/infra level, not form level

`docs/appchanges.txt` is empty (0 bytes) and contributed nothing.

Important: SPEC is *not* a superset. Ventilation fields, imaging/photo capture, ESI triage, the follow-up sheet, and roughly half the AOD timings appear only in N1 – N4 and CHAT . Building from the docx alone would miss them.

Reading the change tables

Marker	Meaning
ADD	Field does not exist today
KEEP	Exists, unchanged
CHANGE	Exists, but type/options/label change
REMOVE	Delete from the instrument
MOVE	Same field, different position in the flow
▲	Sources conflict, or a clinical decision is needed — see §10

2. The headline change

The instrument being edited is the **ICU / APACHE-IV register** (`src/domain/questionnaire/icu-apache.v1.json`, 49 fields). Every field the sources name — CR number, serial number, pre-ICU LOS, emergency surgery before ICU, APACHE, APS — is a field in that file.

The ICU/APACHE-IV instrument is being repurposed into an Emergency Department instrument. It stops being an ICU admission form scored with APACHE IV, and becomes an ED arrival form scored with SOFA.

This is a rename and a re-scope, not a tweak:

- APACHE IV comes out of the ED flow entirely — "Don't use APACHE 4, doesn't work in emergency" CHAT . It moves to a **follow-up sheet** filled after stabilisation.
- APS is dropped — "APS not for ED" CHAT , "No APS" N3 .
- **SOFA replaces it**, with an in-hospital mortality estimate.
- The form is re-sequenced around **who is standing in front of the patient and when**: paramedics fill GCS at the door, so GCS moves to page 1.
- Data entry shifts from typing to **photographing** — the transmission report, the QR code on the OP card, lab strips, the patient monitor, the ventilator, the ECG.

Practical consequence: the ED/ICU toggle in App.tsx currently distinguishes two registers. After this change the "ICU" register is an ED register. Decide early whether it **replaces** the existing ED triage register (triage-abcde.v1.json) or sits alongside it as a second ED instrument — see §10.1. This decision gates a lot of the work below.

3. Flow and page order

Currently four screens. The new sequence, with the reason each move is required:

Page	Contents	Who fills it	Why here
1 — Arrival	Photo of transmission report + QR scan → auto-populate · CR number · admission details · patient category · GCS	Paramedic / senior paramedic / junior PG	"GCS → on arrival it's done" N3 ; "Senior paramedic or younger pg fills up" CHAT . GCS is on page 4 today, which is wrong — the person who observes it has left by then.
2 — Physiology	Febrile/afebrile · MAP, SBP, DBP · arrival SpO ₂ · HR · RR · FiO ₂ · S/F ratio · blood gas · ventilation	Nurse / PG	Monitor is at the bedside; photograph it.
3 — Labs & assessment	Labs (from strip photo) · urine status · ESI triage · vasopressors · provisional diagnosis · SOFA · imaging/photos	PG / doctor	Lab strips arrive later than vitals.
4 — Outcome	Patient outcome · interventions done in ED · disposition	Doctor, at disposition	Already correct — do not move it onto the entry form.
Separate	Ambulance Offloading Data (AOD) + dashboard	Nurse / registration	New module, §8
Separate	Follow-up sheet (APACHE IV, definitive care)	Ward/ICU team, ~day 2	New module, §9

"Definitive care happens about a day after admission into ED" CHAT — this is why the follow-up sheet must be a separate entry point on the board, not a page of the arrival form.

4. Page 1 — Arrival

4.1 Photo capture and auto-population SPEC §1 , N1 , CHAT

The first screen is the **transmission report**. Nothing is typed that a camera can read.

#	Requirement	Source
1	ADD — Start the form with a photo of the admission / transmission report	CHAT "Start with a photo of the admission page"
2	ADD — Scan the QR code printed on the OP card ; auto-populate patient demographics from it	SPEC §1 , N1 "QR (scan the OP card)", CHAT "Has QR code"
3	ADD — Auto-populate lab and blood-gas values from a photo of the paper data strip	CHAT "Auto populate as well. From strip of paper"
4	Every auto-populated field must remain editable , and visibly marked as machine-read vs. human-entered	Implied — see §10.4

▲ One known gap, from the sample photos: "The starting id of 33101 isn't there. Rest is available on these data strips" CHAT . The patient identifier is **not** on the lab strip — only the values are. The CR number must come from the QR/OP card or be typed. Strip OCR cannot be the sole identity path, so **auto-populated labs must be bound to an already-identified encounter**, never used to create one.

Note the earlier decision this reverses: the app README records "Barcode CR scan was cut deliberately." It is now back in scope, as QR rather than barcode.

4.2 Admission fields SPEC §2 , N1 , CHAT

Against `icu-apache.v1.json` → `admission` group as it exists today:

Field (<code>TinkId</code>)	Action	Detail
<code>cr_number</code>	KEEP / MOVE	The identifier everything hangs off. Sits in Admission, auto-filled from QR capture. "CR number → identification (company registration)" N1
<code>patient_name</code>	KEEP	Unchanged
<code>phone_number</code>	KEEP	Unchanged
<code>age_value</code>	KEEP	Unchanged
<code>age_unit</code>	CHANGE	Options become years, months only. Remove "days" — "Remove days (Adult Emergency)" N1
<code>sex</code>	KEEP	Unchanged
<code>origin</code>	CHANGE	Reduce from 5 options to 2: Direct admission • Other hospital . SPEC §2 , N1 . ▲ CHAT words it "referral or direct admission" — same two-way split, different label. Recommend " Direct admission / Referred from another hospital ".
—	ADD	Other hospital details — conditional on <code>origin = Other hospital</code> . Placement deliberately undecided: "not necessarily on the first page" SPEC , "(not on admission pages)" N1
—	ADD	Days spent in the other hospital — "how many days in other hosp" N1 . This is the replacement for pre-ICU LOS: "Pre ICU data — should be pre NIMS hospitalization" CHAT
—	ADD	Where is the patient coming from + address — "Where is he coming from → imp info", "Address" N1 , CHAT
<code>icu_serial_no</code>	REMOVE	"Remove serial no. Only CR no" CHAT , SPEC §2
<code>pre_icu_los_days</code>	REMOVE	Superseded by days-in-other-hospital above
<code>emergency_surgery</code>	REMOVE	Superseded by Patient category, §4.3
<code>icu_admission_date</code>	CHANGE	Relabel to ED arrival date/time; reconcile with the AOD timings module (§8) so the same timestamp is not captured twice

4.3 Patient category — new SPEC §3, N1, CHAT

ADD a single required classification, placed before the clinical sections. Every arriving patient is exactly one of:

- Surgical emergency
- Medical emergency
- Trauma emergency

N1: "Has patient any → medical / surgical / trauma". CHAT: "Medical/ surgical or trauma case", "Has patient undergone any surgical intervention".

This field is likely to drive downstream branching (trauma → log roll and exposure; surgical → intervention capture). Worth building as a proper branch key rather than a flat choice.

4.4 GCS — moved earlier SPEC §4, N3, CHAT

Requirement	Detail
MOVE	Out of the <code>chronic_neuro</code> group on page 4 → onto page 1, filled on arrival
CHANGE	Eye, verbal and motor all become number-selection dropdowns — "Drop-down list box — only numbers" CHAT
KEEP	Ranges: Eye 1–4 · Verbal 1–5 plus T (intubated) · Motor 1–6. N3 writes "1–4 → eye, 5 → verbal → 6 (intubated), 6 → motor"
—	Filled by paramedics / PG staff — N3 "(paramedics / PG people)"

The existing ED instrument already models verbal correctly (choice capped at 5 + T); carry that model over rather than reinventing it. `gcs_total` stays a calculated read-only field.

5. Page 2 — Physiology, blood gas and ventilation

5.1 Physiology SPEC §5, N1, CHAT

Field	Action	Detail
temperature + temperature_unit	CHANGE	Replace the numeric + unit pair with a binary: Febrile / Afebrile . Numeric value not needed. SPEC §5, N1 "Temp → Febrile / Afebrile", CHAT "Page 2 — Febrile or afebrile"
map_measured	KEEP	MAP retained
—	ADD	Systolic BP
—	ADD	Diastolic BP
N1 : "MAP, SP, DP → all three". All three are captured, not derived from each other.		
hr	KEEP	Heart rate
rr	KEEP	Respiratory rate
—	ADD	Arrival SpO₂ — "Arrival spo2" CHAT, N1. Label it <i>arrival</i> explicitly; it is a specific timepoint, not a running observation
fio2	KEEP	FiO ₂
—	ADD	S/F ratio (SpO ₂ / FiO ₂), calculated read-only. Replaces P/F as the default — "SF ratio and not PF ratio" CHAT, SPEC §5
—	ADD (conditional)	P/F ratio — N1 qualifies it: "S/F ratio · P/F ratio → depending on patient". Show P/F only where an arterial gas exists, i.e. ventilated patients (§5.2). See ▲ §10.2
—	ADD	Patient monitor photo — "Patient monitor vitals, added" N1, "Patient monitor pictures" N2

5.2 Blood gas SPEC §5, N2, CHAT

Requirement	Detail
CHANGE	Default gas is VBG, not ABG SPEC §5
ADD	A source selector on pO₂: ABG or VBG — N2 "PO ₂ (option) → ABG / VBG", CHAT "Po2 — give option abg or [v]bg". The value is meaningless without knowing which
CHANGE	Full ABG panel only for mechanically ventilated patients — "Full abg for mech ventilation ONLY" CHAT. The existing po2 / pco2 / arterial_ph fields become conditional on mechanical_vent = Yes
ADD	Lactate and base deficit , read from the gas panel — N2 "Lactate → ABG/VBG (take from that)", CHAT "Serum lactate, Base deficit, From abg vbg itself". ▲ Conflicts with SPEC §6, which drops lactate. See §10.3
ADD	Auto-populate the gas panel from a photo of the strip — N2 "Blood gases info → Auto populate"
ADD	Sample strip photo (ABG/VBG) — N2

5.3 Ventilation — new, missing from SPEC N2, CHAT

Not in the docx at all. Conditional on mechanical_vent = Yes :

Field	Action	Source
Ventilator mode	ADD — free text (N2 says "ventilator mode → write"; consider a coded list with an "other" escape)	N2 , CHAT "What mode of mech ventilation"
Tidal volume	ADD	N2 "Tidal volume: Add", CHAT
PEEP	KEEP — already exists	CHAT "Fio2, tidal volume, peep"
FiO ₂	KEEP — already exists	CHAT
Ventilator monitor photo	ADD	N2 , CHAT "Picture of the monitor of ventilator"

6. Page 3 — Labs, triage, drugs, diagnosis

6.1 Labs SPEC §6, N2, CHAT

The governing principle, in the clinicians' own words: "Emergency → excretory functions (creatinine / urea). Albumin is not included" N2, and "Less focus on synthetic functions" CHAT. Liver synthetic function is not an ED concern.

Field	Action	Detail
sodium	KEEP	"Correct as it stands" SPEC §6
creatinine	KEEP	Excretory — explicitly retained N2
urea	KEEP	Excretory — explicitly retained N2
glucose	KEEP	Not mentioned for removal
hematocrit	KEEP	Not mentioned for removal
wbc	KEEP	Not mentioned for removal
urine_output_24h	CHANGE	Replace the 24-hour numeric with a 3-way choice: Oliguric • Anuric • Normal . "Nothing time-based — no waiting 24 hours" SPEC §6 ; N2 shows "No UO" struck out and replaced by "Anuric / Oliguric (less output) / Normal"; CHAT "Is the patient anuric or not?"
albumin	REMOVE	Synthetic function, not needed
bilirubin	REMOVE	Synthetic function, not needed. ▲ Required by SOFA — see §10.3
lactate	MOVE	Out of Labs; read from the blood gas panel instead (§5.2)
base_deficit	MOVE	Same — from the gas panel
—	ADD	Generic labs page — "Labs — make a generic page" CHAT. A schema-driven lab section that can take new analytes without a code change, rather than a fixed list

6.2 Triage — ESI N2, CHAT

ADD ESI category **1–5** to this instrument — N2 "Triage → ESI 1, 2, 3, 4, 5 mostly".

The existing ED triage instrument already has a working esi field with the right options (triage-abcde.v1.json → triage group). Reuse that definition rather than authoring a second one; the board already colour-codes and sorts on ESI.

6.3 Vasopressors SPEC §8, N3, CHAT

Field	Action	Detail
vasopressor_used	CHANGE (label)	The yes/no must read "In use, or started within 1 hour of arrival" . "So can't be only in-use or no" CHAT — the current label is wrong and changes what gets recorded
—	ADD	How many — dropdown: 1, 2, 3. SPEC §8, N3 "How many → 1, 2 ... dropdown"
vasopressors_detail	CHANGE	Free text → multi-select naming which agents , count driven by the field above

Agent list (union of SPEC §8, N3, CHAT — all three agree):

1. Norepinephrine
2. Epinephrine
3. Dopamine
4. Dobutamine (written "Dovitamin" in CHAT — transcription of dobutamine)
5. Vasopressin
6. Phenylephrine
7. Methylene blue

6.4 Diagnosis and scoring SPEC §9, N3, CHAT

Field	Action	Detail
diagnosis	CHANGE (label)	"ICU admission diagnosis" → "Provisional diagnosis" (working diagnosis, ED). SPEC §9, N3, CHAT "Provisional diagnosis / Working diagnosis"
apache_iv_score	MOVE	Out of the ED flow → follow-up sheet (§9). "Don't use apache 4 — doesn't work in emergency" CHAT . Note N3 : APACHE 4, not APACHE 2
aps_score	REMOVE	"APS not for ED" CHAT , "No APS" N3
—	ADD	SOFA score — 6 parameters. "They do SOFA scoring (6 parameters) → for ED" N3
estimated_mortality_rate	KEEP / CHANGE	Re-base on SOFA. Label as in-hospital mortality — "In house mortality" CHAT ×2
estimated_los_days	▲ DECIDE	An APACHE-IV output. No source mentions keeping it. Recommend remove from the ED instrument
—	ADD	MDCalc integration for SOFA and mortality — SPEC §9, N3 "MD Calc integrate: SOFA sure", CHAT "MD calc". ▲ See §10.5

6.5 Chronic health — deleted SPEC §7

REMOVE the entire block. Hepatic failure, cirrhosis, metastatic carcinoma, lymphoma, leukaemia/ myeloma, immunosuppression, AIDS, chronic renal failure on haemodialysis — all out. These are APACHE comorbidity weights and have no role in an ED arrival form.

Note this empties the chronic_neuro group: GCS leaves for page 1 (§4.4), vasopressors move to page 3 (§6.3), everything else is deleted. **The group ceases to exist.**

6.6 Imaging and photos N2, CHAT

Requirement	Detail
ADD	ECG photos — "Put ECG photos here also" N2
ADD	Patient monitor photos — N2
ADD	Ventilator monitor photos — N2
ADD	ABG/VBG sample strip photos — N2
DO NOT ADD	No X-ray report field. "Radiology reports are generated only for CT and MRI. Not for chest x ray" CHAT ; "No X-ray report available" N4 . A chest film exists as an image only — there is no report to transcribe, so do not build a field that will sit permanently empty
ADD	CT / MRI report capture (text or photo) — these do exist

Image quality is a stated requirement, not a nice-to-have: "*photos aren't blurry, pretty ad inputs, filters checking*" IDEAS . Blur/legibility validation belongs at capture time — a blurred strip that fails OCR at 3am is a lost record. See §11.

7. Page 4 — Outcome SPEC §11, N4, CHAT

Field	Action	Detail
—	ADD	Patient outcome — what happened to the patient afterwards. SPEC §11, N4
—	ADD	Interventions done in emergency — SPEC §11, N4, CHAT "What interventions are done in the ED". CHAT adds "Should be on the case sheet" — i.e. this is transcribed from the paper case sheet
—	ADD	Did the patient improve or not — CHAT . A distinct question from disposition; capture separately
—	ADD	Time patient was shifted out — CHAT ; feeds the AOD dashboard (§8)
icu_outcome	CHANGE	Relabel the existing outcome choices for ED disposition rather than ICU discharge

Keep the existing architectural decision: outcome stays on its **own worklist entry point**, not on the entry form. The README records that outcome-as-a-field decayed to 0% fill within eight weeks in the Google Form era. Nothing in these sources asks to reverse that.

8. New module — Ambulance Offloading Data (AOD) SPEC §10, N4, CHAT

"Overboarding in emergency is directly related to mortality" CHAT "ED overboarding → what time he was in trolley → lean methodology" N3

This is the module with the clearest research purpose: it exists to measure **delay**. Each timestamp is only useful as one end of an interval.

8.1 Timestamps to capture

All entered as times. SPEC §10 lists four; N4 and CHAT add the rest.

#	Timestamp	Source	Entered by
1	Time paramedics picked the patient up	SPEC §10	Paramedic
2	Time of ambulance arrival at ED	SPEC §10, N4, CHAT	Registration
3	Time patient taken inside / offloaded	N4, CHAT "Time of pt taken inside", "Offloaded"	Registration
4	Time of admission / entry register	N4 "entry register", CHAT	Registration
5	Time patient was put in trolley	CHAT "Time in trolley", N3	Nurse
6	Time first seen by a doctor	SPEC §10 "Doctor checking time", N4, CHAT "Doc first seen"	Doctor
7	Time blood sample was sent	N4 "Blood sample sent (nurse side)", CHAT	Nurse
8	Time blood report came back	SPEC §10, N4, CHAT "Time when blood reports are sent"	Nurse
9	Time seen by the definitive care doctor	N4 "Definitive care doctor", CHAT	Doctor
10	Time patient was shifted out of ED	CHAT	Nurse

8.2 Derived intervals — the actual output

These are what the module is for. Compute and display them; do not make anyone subtract times by hand.

Interval	From → To	What it measures
Ambulance transit time	1 → 2	"How much time he was in ambulance" N4
Offload delay	2 → 3	"How much time to take inside" N4 — the module's namesake
Door-to-doctor	3 → 6	Triage responsiveness
Nurse delay	7 → 8	Explicitly named: "Doc seen — time of blood report → nurse delay" CHAT
Trolley / boarding time	5 → 10	ED overboarding — the mortality link
Door-to-definitive-care	3 → 9	Typically ~1 day CHAT
Total ED length of stay	2 → 10	Throughput

8.3 Entry mode ▲

The sources contradict themselves on this point, in the same message run:

- "Manually fill this" CHAT
- "Needs to be manually entered" CHAT
- "Automate into the app" N4

Recommended resolution: capture each timestamp with a **one-tap "now" button** at the point of care — the staff member taps when the event happens, and the app stamps the time. That is manual entry (a human asserts the event occurred) with no typing and no clock arithmetic. Reserve full automation for timestamps the app already owns, such as the moment the record is created. Confirm with Dr Sabheeha.

8.4 AOD dashboard N4, CHAT

A dashboard is explicitly requested, listing:

- Ambulance arrival
- Offloaded
- Doctor first seen
- Blood report time
- Blood sample sent
- Definitive care doctor

Plus a **date and time filter**: "Dropdown clock (date & time)" N4 .

Build it against the derived intervals in §8.2 rather than the raw timestamps — a table of ten clock times tells no one anything, whereas median offload delay by hour of day is the finding.

9. New module — Follow-up sheet N3 , CHAT

APACHE IV does not belong in the ED, but it is still wanted — *later*.

Requirement	Detail
ADD	A follow-up sheet, filled after stabilisation — "APS, APACHE data after stabilizing" N3 , "Page 4 needs to be filled — aps and apache scores need to be done... done after stabilization" CHAT
ADD	APACHE IV scoring lives here — "Follow up sheet where apache 4 can also be filled" CHAT . APACHE 4, not APACHE 2 N3
ADD	Definitive care record — "Sample of how the definitive care happens about a day after admission into ED" CHAT
ADD	Its own board entry point and worklist, like the outcome sheet

This is where the chronic-health block should be re-homed rather than deleted outright. APACHE IV cannot be scored without the comorbidities being removed in §6.5. They are wrong on the ED arrival form and necessary on the follow-up sheet. Preserve the field definitions when deleting them from the ED instrument.

10. Conflicts and decisions needed ▲

These must be resolved before implementation. Each one changes what gets built.

10.1 Does this replace the existing ED register, or sit beside it?

The app already has an ED instrument (`triage-abcde.v1.json` , 58 rules, ABCDE assessment) and an ICU instrument. These changes convert the ICU instrument into an ED one — which leaves **two ED instruments** with overlapping fields (CR number, age, sex, GCS, SBP/DBP, SpO₂, RR, ESI). IDEAS names this problem directly: "*duplicate ideas... patient ID standardization... what's the good way to do it*".

Three options: 1. **Merge** — one ED instrument, the ABCDE assessment becoming a section of it. Cleanest data, largest job. 2. **Keep both** — ABCDE for walk-ins, the new form for ambulance/resus arrivals, keyed on CR number so they join. 3. **Replace** — retire ABCDE.

Recommendation: option 2 for the pilot — it ships fastest and does not disturb a working instrument — **with CR number enforced as the join key from day one**, so option 1 remains reachable later. Needs Prof. Ashima Sharma's call.

10.2 SOFA needs fields this spec removes

SOFA's six parameters are: respiration (PaO₂/FiO₂), coagulation (**platelets**), liver (**bilirubin**), cardiovascular (MAP + vasopressors), CNS (GCS), renal (creatinine + urine output).

The changes above:

- **Remove bilirubin** (§6.1) — SOFA's liver component
- **Never add platelets** — SOFA's coagulation component; not in the current form and not requested by any source
- **Replace P/F with S/F** (§5.1) — SOFA's respiratory component is defined on PaO₂/FiO₂
- **Replace numeric urine output with a 3-way category** (§6.1) — SOFA's renal component uses mL/day

MAP, vasopressors, GCS and creatinine are all fine. **Four of the six components are affected.** Options:

1. Bring back bilirubin and add platelets, ED-only, justified as SOFA inputs.

2. Compute a **partial SOFA**, with unscored components shown explicitly as unscored — never silently zero. *(The app's existing Kleene three-valued evaluator already treats blank as UNKNOWN rather than zero; this is the natural fit.)*
3. Use a published modified SOFA that accepts S/F and categorical urine output, and state which variant on the export.

Recommendation: option 2 for the pilot, with the score labelled as partial — but this is a clinical decision, not an engineering one. Raise it before the build, because a SOFA score that silently treats a missing bilirubin as normal is a wrong number presented as a right one, and it will propagate into the research output.

10.3 Lactate — the sources disagree

SPEC §6 says lactate is "not needed right now". N2 and CHAT both say to take lactate and base deficit from the ABG/VBG.

Resolution taken in this document: both are satisfied — remove the *separate serum lactate lab field* (SPEC is right that no extra test is ordered), and surface lactate and base deficit from the **gas panel**, where they already come for free. Recorded as §5.2. Confirm.

10.4 Provenance of auto-populated values

Not raised by any source, but unavoidable once OCR is in: a sodium read by a camera and a sodium typed by a PG are not equally trustworthy, and the app's whole design premise is entry-time data quality. Every auto-populated field needs a provenance flag (`qr` / `ocr` / `manual`), visible in the UI and carried into the export. Cheap now, retrofit-hostile later.

10.5 "Integrate MDCalc" — what does it mean?

Requested three times. MDCalc has no public calculation API, so the literal reading may not be buildable. Three interpretations:

1. **Implement SOFA in-app** and validate the output against MDCalc's calculator. *(Recommended — keeps the app offline-capable, which matters for an ED.)*
2. **Deep-link out** to MDCalc with values pre-filled, clinician copies the score back. Breaks offline use and loses the audit trail.
3. **Licensed API access** — commercial arrangement with MDCalc; long lead time.

Recommendation: option 1, with the SOFA rules held as data in the rulebook so the scoring is versioned and auditable like everything else. Confirm what was meant.

10.6 Smaller opens

- **Estimated ICU LOS** (§6.4) — keep or drop? No source mentions it. Recommend drop.
- **Other-hospital details placement** (§4.2) — deliberately left open by two sources. Recommend a conditional section revealed by `origin`, on page 1 — it is admission data and belongs with admission.
- **Ventilator mode** (§5.3) — free text as written, or a coded list? Free text is what produced the 65-spelling `Log Roll` field. Recommend a coded list with an "other" escape.

11. Platform and infrastructure IDEAS

From `duplicate ideas evidence.txt` — a different altitude from the form changes, and mostly not yet scheduled. Recorded here so it is not lost.

Item	Note	Status
Flutter deployment	" <i>deploy it on flutter</i> " — the app is currently React/Vite (web). A rewrite, not a port. Assess whether a PWA meets the need first	⚠ Major — needs a decision
Basic login ID	" <i>basic login id with flutter app</i> ". The README confirms <code>USER</code> in <code>App.tsx</code> is still a placeholder, and the audit found 35 collectors sharing one form login	Open, high priority
DPDP compliance	" <i>figure out regulatory DPDP store and retrieve it safely</i> " — India's Digital Personal Data Protection Act. The two-store split (<code>encounters</code> holds no identifiers; <code>identifiers</code> keyed separately) is the right foundation; encryption at rest and a retrieval policy are not done	Open, blocks real patient data
Patient ID standardisation	" <i>patient ID standardization... tailor made exactly</i> ". CR number is the key (§4.2); enforce format validation	Partly covered
Photo quality validation	" <i>photos aren't blurry, pretty ad inputs, filters checking</i> "	Open — see §6.6
Multi-parameter monitor integration	" <i>multiparameter monitoring system, systolic bp diastolic bp</i> ". Photo capture (§5.1) is the interim answer; a direct feed is the eventual one	Future
RFID patient tracking	" <i>RFID based tracker on the patients · tag on the hand · prox sensor clocks patient moving across it</i> ". This is the automated answer to the AOD timestamps in §8 — proximity sensors would stamp offload and movement times without anyone tapping	Future — but note the direct link to §8.3
Departmental scope	" <i>emergency, orthopedics, lots of departments... flexibility on the protocol they are following</i> ". The instrument should not hard-code ED-only assumptions	Design constraint
Teaching-hospital context	" <i>NIMS is a teaching hospital for MD</i> " — data entry is largely by rotating PGs, so the form must survive high staff turnover and minimal training	Design constraint

12. Where each change lands in the code

Change	File(s)
All field add/remove/change (§4–§7)	<code>src/domain/questionnaire/icu-apache.v1.json</code> — rename to an ED-named instrument, bump <code>version</code> , mint a new <code>url</code>
Page order / sequence (§3)	<code>src/domain/questionnaire/icu-adapter.ts</code> , <code>src/ui/IcuEncounterScreen.tsx</code>
Removed/changed fields' validation	<code>src/domain/rulebook/icu.rules.v1.json</code> — 35 rules; every rule referencing a removed field must be retired
S/F ratio, SOFA total, derived intervals	<code>src/domain/derived.ts</code> — calculated, never zero-filled
SOFA scoring (§6.4, §10.2)	New rulebook entries + <code>src/domain/derived.ts</code> ; keep scoring as data, not code
QR scan / OCR / photo capture (§4.1, §6.6)	New — camera permissions, image store. Note <code>db.ts</code> currently holds no blobs
Photo storage & identifier separation	<code>src/data/db.ts</code> — images of a transmission report are identifying ; they belong in the <code>identifiers</code> store, not <code>encounters</code>
AOD module + dashboard (§8)	New screens; new questionnaire artifact; new board entry point
Follow-up sheet (§9)	New questionnaire artifact (re-homing the APACHE IV + chronic health fields); new worklist
ESI on this instrument (§6.2)	Reuse the <code>esi</code> item from <code>triage-abcde.v1.json</code>
Export	<code>src/data/export.ts</code> — new fields, new version stamps, provenance flags (§10.4)
Tests	<code>tests/icu-workflow.test.ts</code> , <code>tests/icu-store.test.ts</code> , <code>tests/defect-catalog.test.ts</code>

Versioning: the app stamps every saved record with a questionnaire version and a rulebook version. This is a breaking schema change — bump both, and do not mutate the existing artifacts in place, or previously saved records become unreadable against their own schema.

13. Suggested build order

#	Work	Why here	Rough size
1	Resolve §10.1 (one register or two) and §10.2 (SOFA inputs)	Both gate schema design; wrong call here is expensive later	Decision only
2	Questionnaire rewrite — all of §4–§7 except photos	Pure data edit, no new infrastructure; unblocks everything	Small–medium
3	Rulebook reconciliation + tests green	Removed fields leave dangling rules	Small
4	Flow re-sequencing, GCS to page 1	UI only, once the schema is right	Small
5	SOFA + mortality (§6.4, §10.5)	Needs the schema settled	Medium
6	AOD module + dashboard (§8)	Self-contained; highest research value; can run in parallel from step 2	Medium
7	Follow-up sheet (§9)	Self-contained	Small–medium
8	Photo capture + storage (§4.1, §6.6)	New infrastructure; touches the identifier boundary and DPDP	Large
9	QR auto-population	Depends on 8	Medium
10	Strip OCR auto-population	Depends on 8; highest technical risk, lowest certainty of working	Large
11	Auth (§11)	Needed before real patient data, independent of all the above	Medium

Items 8–10 are where the effort concentrates. Everything in 2–7 is achievable without a camera, and delivers a usable ED instrument on its own — worth shipping as a first release rather than waiting on OCR.

14. People and site

Clinical lead	Prof. Ashima Sharma IDEAS
Day-to-day contact	Dr Sabheeha CHAT
Location	128 Millennium Block, 1st floor CHAT
Who fills the arrival page	Senior paramedic or junior PG CHAT , N3
Who fills AOD	Registration and nursing staff N4
Institution	NIMS Hyderabad — teaching hospital, MD programme IDEAS

"Please summarise all fixes point by point, so that Bhavith or you can fix it and get going" — CHAT

Appendix — Change count

Section	ADD	CHANGE	REMOVE	MOVE
Admission (§4.2)	3	3	3	1
Patient category (§4.3)	1	—	—	—
GCS (§4.4)	—	1	—	1
Physiology (§5.1)	5	1	—	—
Blood gas (§5.2)	4	2	—	—
Ventilation (§5.3)	3	—	—	—
Labs (§6.1)	1	1	2	2
Triage (§6.2)	1	—	—	—
Vasopressors (§6.3)	1	2	—	—
Diagnosis & scoring (§6.4)	2	2	1	1
Chronic health (§6.5)	—	—	8	—
Imaging (§6.6)	5	—	—	—
Outcome (§7)	4	1	—	—
Form totals	30	13	14	5

Plus three new modules (AOD + dashboard, follow-up sheet, photo/OCR capture), six open decisions (§10), and ten platform items (§11).